


**Figure 2.6 Scales can also be used to map numerical values to color bands, to make it easier to denote values using a color scale.**

The code to create this ramp is the same, except for the reference to colors in the `range` array:

```
var newRamp = d3.scale.linear().domain([500000,13000000]).range(["blue",
"red"]);
newRamp(1000000);
newRamp(9000000);
newRamp.invert("#ad0052");
```

**Returns "#ad0052"**

**The invert function only works with a numeric range, so inverting in this case returns NaN.**

**Returns "#0a00f5", allowing you to draw a city with population 1,000,000 as dark purple**

You can also use `d3.scale.log()`, `d3.scale.pow()`, `d3.scale.ordinal()`, and other less common scales to map data where these scales are more appropriate to your dataset. You'll see these in action later on in the book as we deal with those kinds of datasets. Finally, `d3.time.scale()` provides a linear scale that's designed to deal with date datatypes, as you'll see later in this chapter.

### BINNING: CATEGORIZING DATA

It's useful to sort quantitative data into categories, placing the values in a range or "bin" to group them together. One method is to use quantiles, by splitting the array into equal-sized parts. The quantile scale in D3 is, not surprisingly, called `d3.scale.quantile()`, and it has the same settings as other scales. The number of parts and their labels are determined by the `.range()` setting. Unlike other scales, it gives no error if there's a mismatch between the number of `.domain()` values and the number of `.range()` values in a quantile scale, because it automatically sorts and bins the values in the domain into a smaller number of values in the range.

The scale sorts the array of numbers in its `.domain()` from smallest to largest and automatically splits the values at the appropriate point to create the necessary categories. Any number passed into the quantile scale function returns one of the set categories based on these break points.

```
var sampleArray = [423,124,66,424,58,10,900,44,1];
var qScale = d3.scale.quantile().domain(sampleArray).range([0,1,2]);
qScale(423);
qScale(20);
qScale(10000);
```

**Returns 2**

**Returns 0**

**Returns 2**